Linuxzoo.net

Minimize

User: vinay mummadi (FULL)
Account: 40807459@live.napier.ac.uk
Registered Account
Account Links
Machine State: **RUN** Refresh in: 0:03.
Time left: 39 mins
Boot progress: complete

**control** | *connect* | *stats* | *useful*

Switch On | Nice Switch Off
Pull Out Power | Leave Queue

Kali 2024-4
- fresh linux install image
- use previous image (if available)

# Gentle Revision

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## Objectives

To reset all the check buttons from a previous attempt click here

# Question 1 Access

You first need to start the Kali Virtual Machine (VM). Make sure this is a Kali 2024-4 VM and not some other version, or a different operating system. You then need to log into the VM remotely using a variety of methods. You should try to use the graphical interface VNC, and also SSH.

To connect using VNC, use the CONNECT tab in the control window, and click on JavaScript VNC. Similarly for SSH you can use the JavaScript SSH link or SSH direct using putty.exe or some other SSH client.

The username is "kali" and the password is "kali".
The terminal is started within the graphical interface by clicking on the terminal application in the main menu.

Check Tests: Ready

| | |
|---|---|
| Logged in via VNC | PASS |
| Running a terminal in VNC | PASS |
| Running a non-vnc connection (ssh) | PASS |

# Question 2 Create and Navigate Directories

The "cd" command changes your current directory. On its own it changes it to your HOME directory, otherwise you need to specify a parameter which either is an absolute directory (such as cd /home) or a relative directory (such as cd dir1 which takes you to dir1 in your current working directory).

Type

```
cd
pwd
```

This will take you to your home directory and then Print Working Directory.

What is your current working directory? : /home/kali

Check Tests: Ready

Correct cwd PASS

Make three directories in your HOME directory. Name these directories "magic", "happy", and "sad".

Check Tests: Ready

Three directories created PASS

Change your working directory into "magic". Create two more directories in this directory called "dir1" and "dir2". If you create them in the wrong place delete them using "rmdir".

Check Tests: Ready

| | |
|---|---|
| Two directories created | PASS |
| Not accidentally created in HOME | PASS |

Change directory into dir2.

What is your current working directory? : /home/kali/magic/dir2

Check Tests: Ready

Correct cwd PASS

Change your current working directory to /home/kali. You can either "cd /home/kali" or "cd ..", as ".." means the directory above the current directory in your directory tree. Check with pwd to make sure you are in the right directory. Then try to delete the data directory using "rmdir magic". What message to you get back?

Directory not empty

Check Tests: Ready

Error message displayed PASS

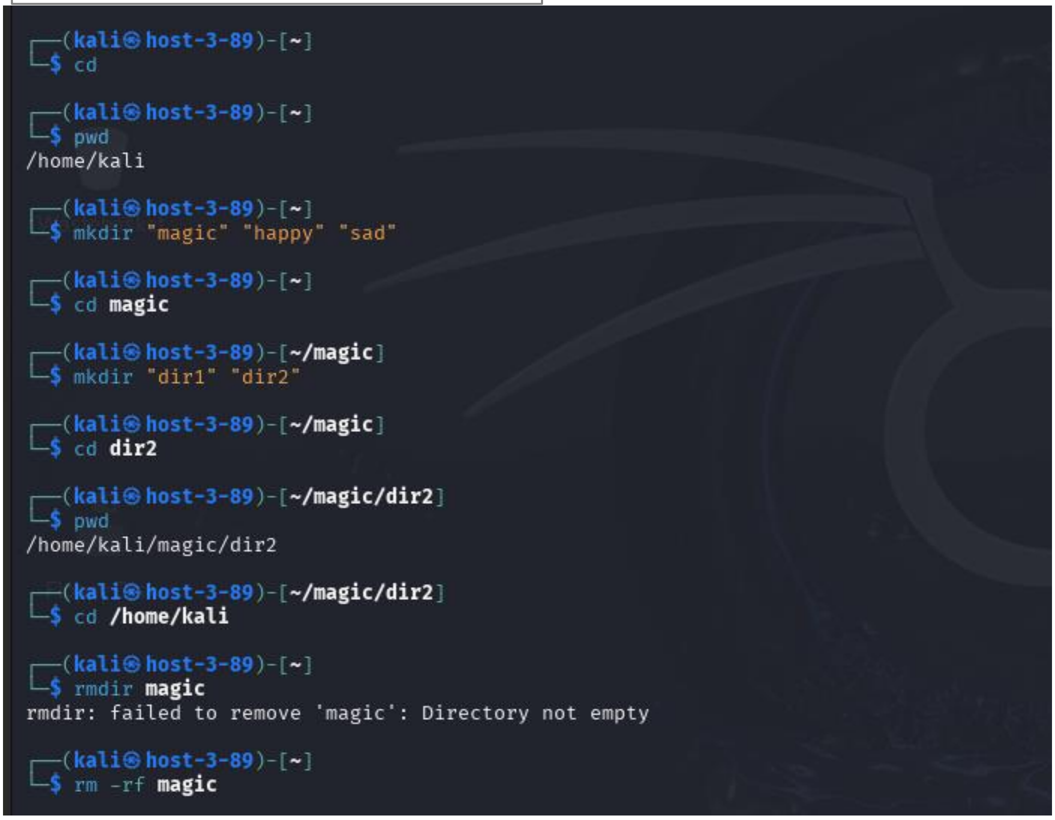
Use the more powerful and dangerous command

```
rm -rf magic
```

This deletes all files and directories in the data directory, including the data directory itself. Obviously getting the parameter wrong means you can lose a lot of stuff in one go!

Check Tests: Ready

Two directories left... PASS



```
┌──(kali㉿host-3-89)-[~]
└─$ cd

┌──(kali㉿host-3-89)-[~]
└─$ pwd
/home/kali

┌──(kali㉿host-3-89)-[~]
└─$ mkdir "magic" "happy" "sad"

┌──(kali㉿host-3-89)-[~]
└─$ cd magic

┌──(kali㉿host-3-89)-[~/magic]
└─$ mkdir "dir1" "dir2"

┌──(kali㉿host-3-89)-[~/magic]
└─$ cd dir2

┌──(kali㉿host-3-89)-[~/magic/dir2]
└─$ pwd
/home/kali/magic/dir2

┌──(kali㉿host-3-89)-[~/magic/dir2]
└─$ cd /home/kali

┌──(kali㉿host-3-89)-[~]
└─$ rmdir magic
rmdir: failed to remove 'magic': Directory not empty

┌──(kali㉿host-3-89)-[~]
└─$ rm -rf magic
```

# Question 3 The ls command

The "ls" command on its own shows you the contents of the current working directory. However it can also take a number of useful parameters.

Filenames which begin with a "." character do not usually appear when you use "ls". These are called "hidden" files, and they are used for things like application configuration and some GUI state information. They are rarely needed so this is the reason they are hidden from normal "ls".

```
┌──(kali㉿host-3-89)-[~]
└─$ ls
a.out        Desktop    Downloads  mbox   overflow   overflow2.c  Pictures  sad        test1     test2.cc  vulnerable.c
dead.letter  Documents  happy      Music  overflow2  overflow.c   Public    Templates  test1.cc  Videos

┌──(kali㉿host-3-89)-[~]
└─$ ls -a
.             .bashrc           Desktop    .face.icon    .java     Music        Pictures                   Templates  .viminfo               .zprofile
..            .bashrc.original  .dmrc      .gnupg        .local    overflow     .profile                   test1      vulnerable.c           .zsh_history
a.out         .cache            Documents  happy         mbox      overflow2    Public                     test1.cc   .Xauthority            .zshrc
.b            .config           Downloads  .hushlogin    .mozilla  overflow2.c  sad                        test2.cc   .xsession-errors
.bash_logout  dead.letter       .face      .ICEauthority .msf4     overflow.c   .sudo_as_admin_successful  Videos     .xsession-errors.old

┌──(kali㉿host-3-89)-[~]
└─$ ls -l
total 152
-rwxrwxr-x 1 kali kali 20400 Feb 27 10:39 a.out
-rw------- 1 kali kali    23 Feb 27 15:19 dead.letter
drwxr-xr-x 2 kali kali  4096 Jan  6  2025 Desktop
drwxr-xr-x 2 kali kali  4096 Jan  6  2025 Documents
drwxr-xr-x 2 kali kali  4096 Jan  6  2025 Downloads
drwxrwxr-x 2 kali kali  4096 Feb 27 18:28 happy
-rw------- 1 kali kali  4791 Feb 27 18:12 mbox
drwxr-xr-x 2 kali kali  4096 Jan  6  2025 Music
-rwxrwxr-x 1 kali kali 17152 Feb 27 10:21 overflow
-rwxrwxr-x 1 kali kali 17152 Feb 27 10:32 overflow2
-rw-rw-r-- 1 kali kali   253 Feb 27 10:31 overflow2.c
-rw-rw-r-- 1 kali kali   243 Feb 26 10:30 overflow.c
drwxr-xr-x 2 kali kali  4096 Jan  6  2025 Pictures
drwxr-xr-x 2 kali kali  4096 Jan  6  2025 Public
drwxrwxr-x 2 kali kali  4096 Feb 27 18:28 sad
drwxr-xr-x 2 kali kali  4096 Jan  6  2025 Templates
-rwxrwxr-x 1 kali kali 20400 Feb 26 11:43 test1
-rw-rw-r-- 1 kali kali   462 Feb 26 11:38 test1.cc
-rw-rw-r-- 1 kali kali   367 Feb 26 12:53 test2.cc
drwxr-xr-x 2 kali kali  4096 Jan  6  2025 Videos
-rw-rw-r-- 1 kali kali   224 Feb 26 16:54 vulnerable.c
```

Use the option "-a" to see the hidden files. View the hidden files in your HOME directory (i.e. /home/kali). The file list starts with "." (which is a directory that you can use to describe the current directory in commands) and ".." (which describes the directory above this one in the tree).

What is the third file or directory (in alphabetical order) which appears when you use "-a" in your HOME directory. You should consider "." as the first file and ".." as the second file. :

.bash_logout

Check Tests: Ready

Third file when using -a PASS

Use the "-l" option to see a long listing of a file. Get a long listing of "/etc/group" and identify its size in bytes. :

1334

Check Tests: Ready

Size of /etc/group PASS

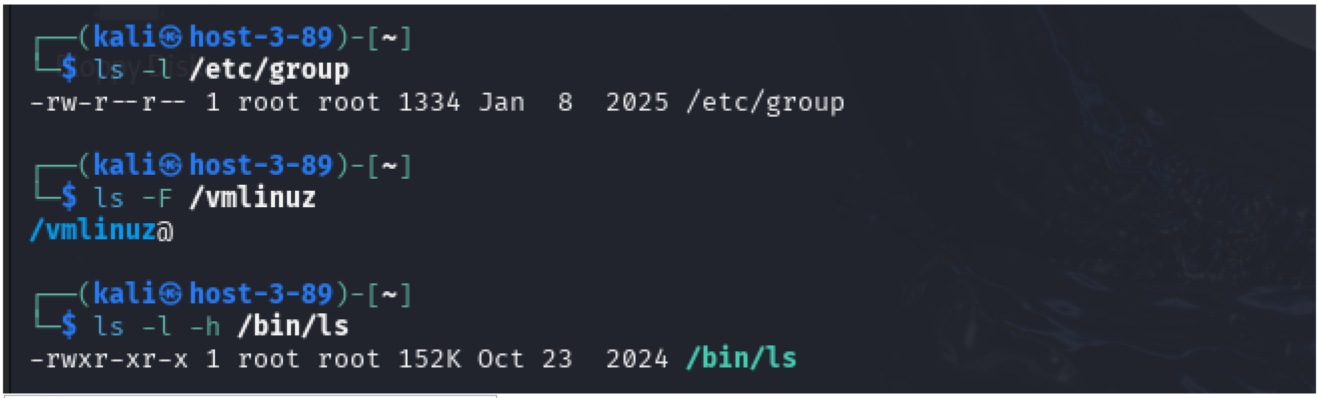
Use the "-F" option to see identification information about files. The "type" appears as a single character at the end of each filename. Use this option and identify the type of /vmlinuz :

@

Check Tests: Ready

Type character of /vmlinuz PASS

Use the "man" command to see the man page for "ls", i.e. "man ls". Use the cursor keys to move around, and find the option which makes ls give its output in human readable form for sizes. Use that in combination with the "-l" option to get the human readable size of "/bin/ls". Include the units, so if in human readable form the size is "10M" type "10M" (case sensitive). :




152K

Check Tests: Ready

Human readable size of /bin/ls PASS

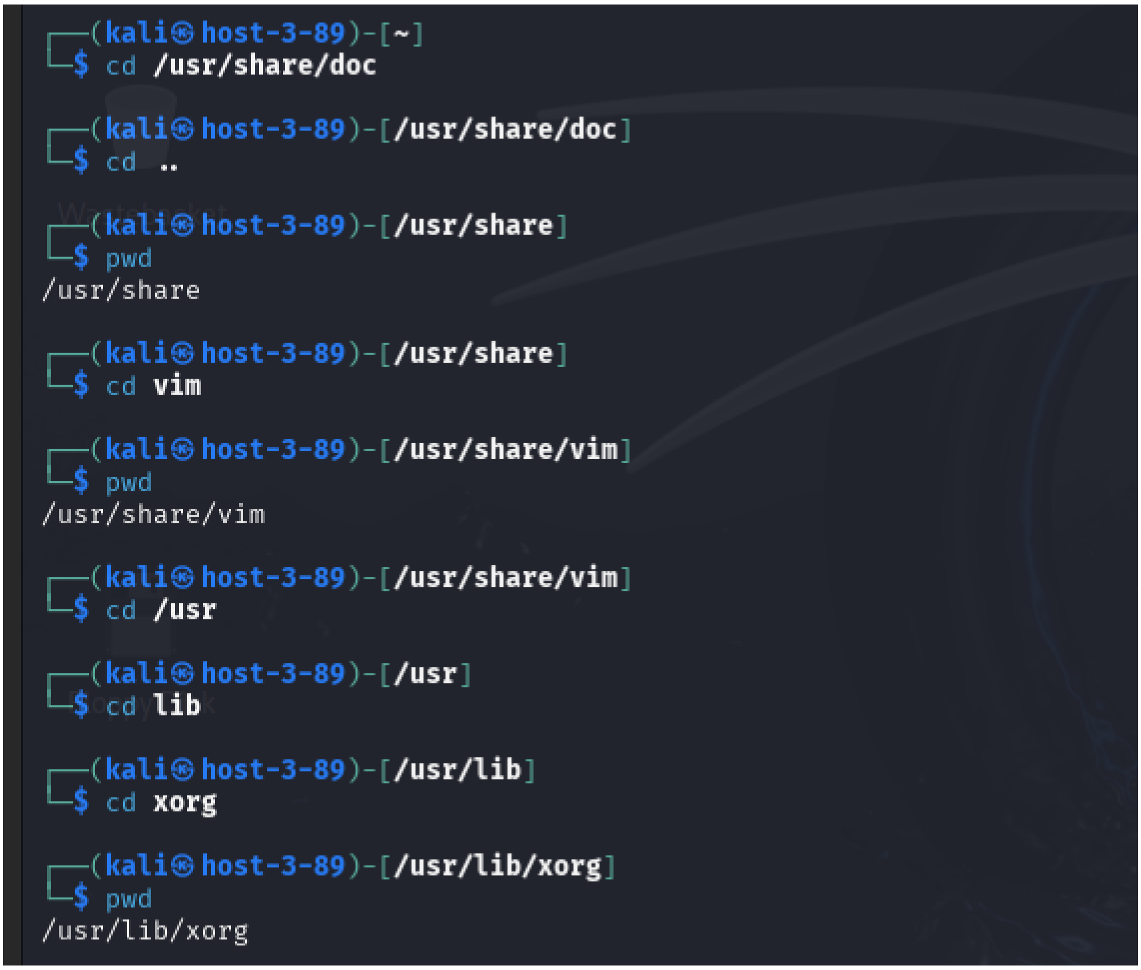
# Question 4 Relative and Absolute

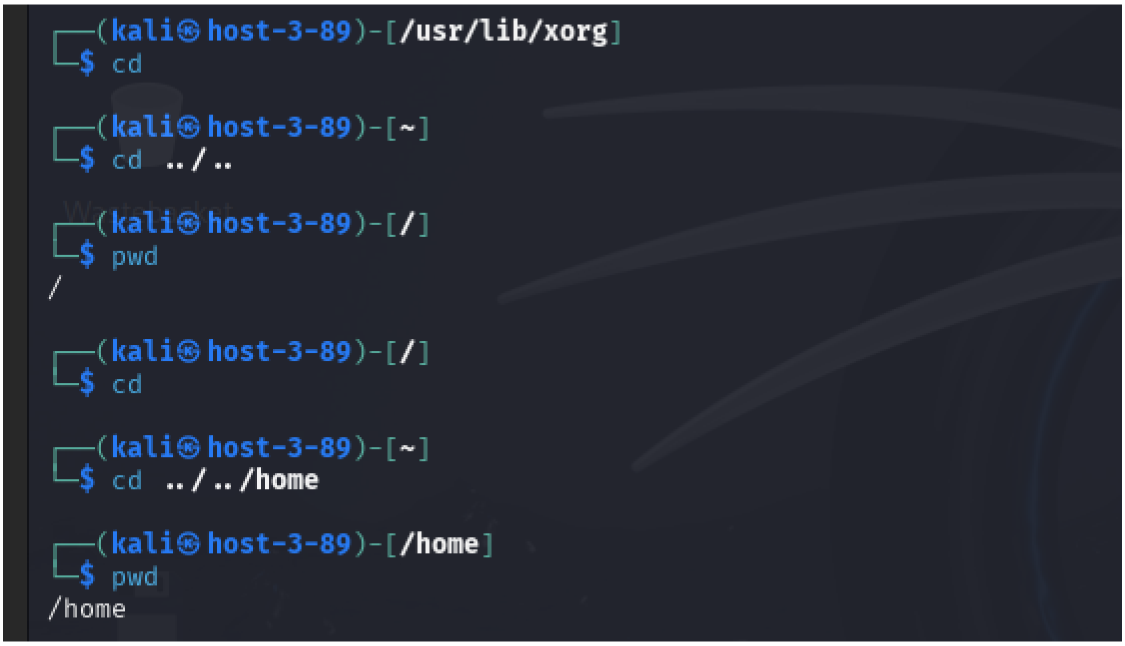
Demonstrate your understanding of relative, absolute, and the use of ".." and "/" and "." in the "cd" command by answering these questions. Try to answer them in your own head, and use the command line only if you are confused. Remember paths beginning with "/" are relative to the top level directory, whereas others are relative to the current directory. ".." takes you up a directory, and things can be seperated with "/" such as "../.." taking you up 2 directories. "." indicates the current directory, and is really only useful where you really need to specify a parameter but you mean to say the current directory. Always supply the SHORTEST solution, remember the answers are CASE SENSITIVE, and if you type spaces in an answerbox where no space should be entered then it will be marked incorrectly.

Do not type the command in unless really stuck. Do these in your head!

Consider these commands:

```
cd /usr/share/doc
cd ..
```



```
pwd
```

What is the current directory: /usr/share

Check Tests: Ready

Correct cwd PASS

Consider these commands:

```
cd /usr/share
cd vim
pwd
```

What is the current directory: /usr/share/vim

Check Tests: Ready

Correct cwd PASS

Consider these commands:

```
cd /usr
cd lib
cd xorg
pwd
```

What is the current directory: /usr/lib/xorg

Check Tests: Ready

Correct cwd PASS

Consider these commands:

```
cd
cd ../..
pwd
```

What is the current directory: /

Check Tests: Ready

Correct cwd PASS

Consider these commands:

```
cd
cd ../../home
pwd
```

What is the current directory: /home

Check Tests: Ready

Correct cwd PASS

Consider these commands:

```
cd /usr/share
cd vim/../../lib/xorg
pwd
```

What is the current directory: /usr/lib/xorg

Check Tests: Ready

Correct cwd PASS

Consider these commands:

```
cd /usr/share
cd /etc/vim
cd ..
pwd
```

What is the current directory: /etc

Check Tests: Ready

Correct cwd PASS

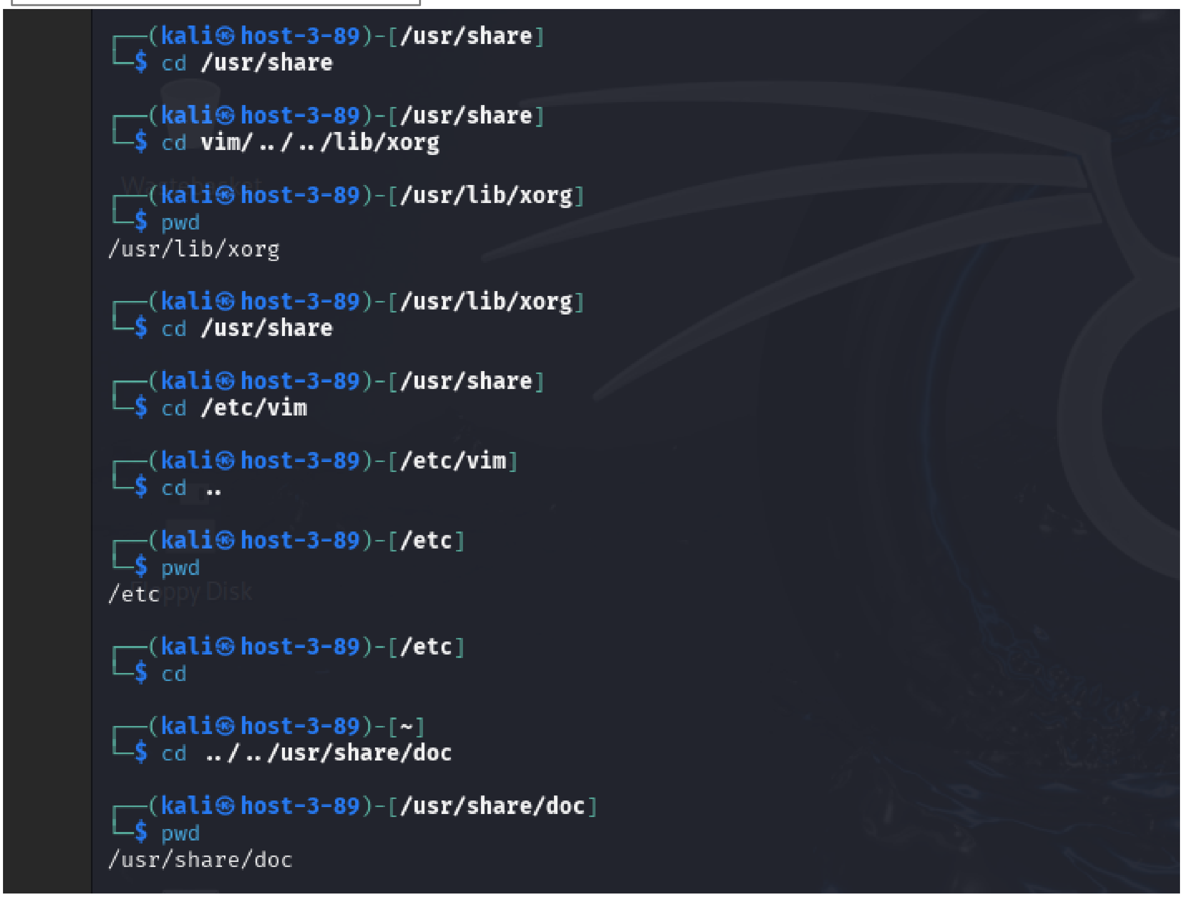
Consider these commands:

```
cd
cd ../../usr/share/doc
pwd
```

What is the current directory: /usr/share/doc

Check Tests: Ready

Correct cwd PASS



Consider these commands and fill in the blank:

```
cd
cd ../ ../usr/local/lib
pwd
```

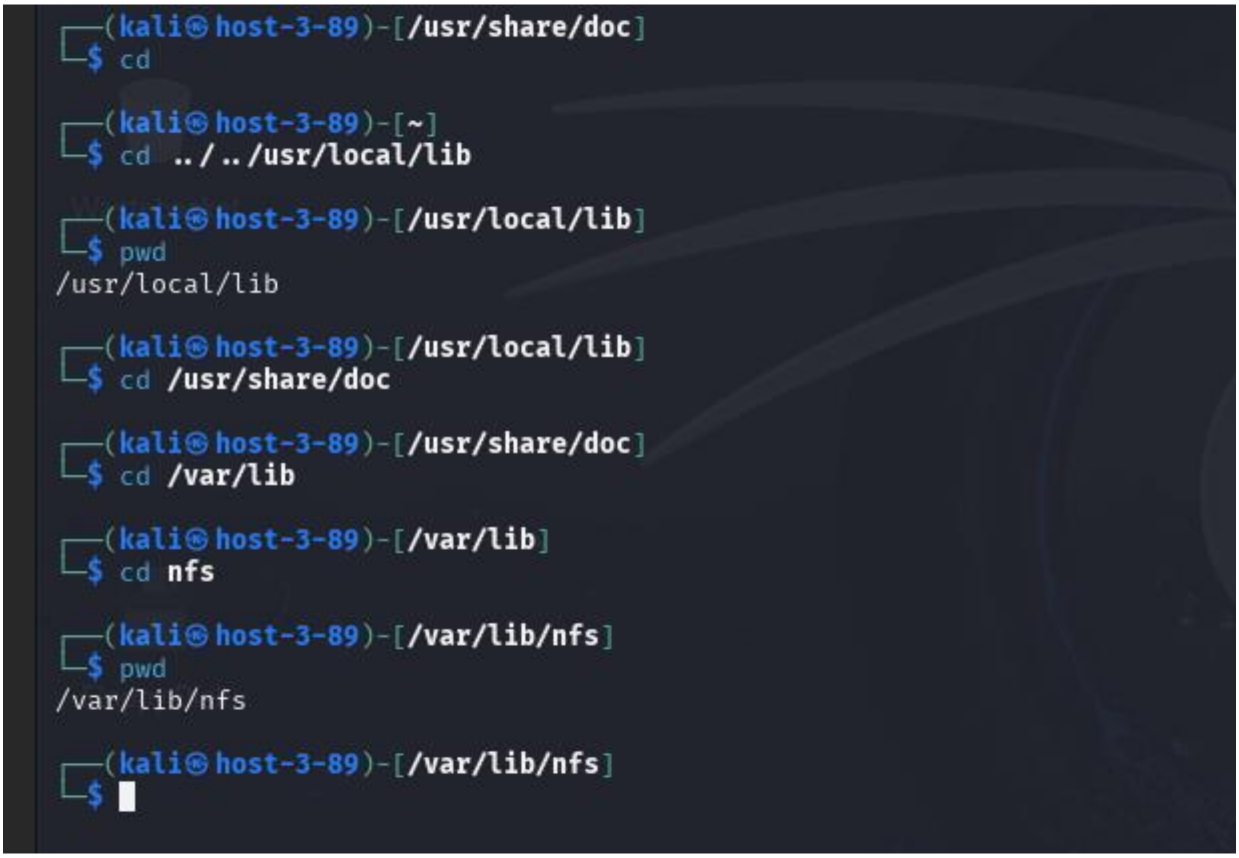
Where the "pwd" command prints "/usr/local/lib".
Note this is an example of a RELATIVE pathname parameter to cd, as the parameter does not start with a "/".

| Check | Tests: Ready |
|---|---|
| | Correct blank PASS |

Consider these commands and fill in the blank:

```
cd /usr/share/doc
cd / var/lib
cd nfs
pwd
```

Where the "pwd" command prints "/var/lib/nfs"
Note this is an example of an ABSOLUTE pathname parameter to cd, as the parameter starts with a "/".



Check Tests: Ready

Correct cwd PASS

Consider these commands and fill in the correct blank using a RELATIVE pathname. It should be the SHORTEST possible solution.

```
cd /usr/share/python
cd ../perl5/Encode
pwd
```

Where the "pwd" command prints "/usr/share/perl5/Encode".

Check Tests: Ready

Correct cwd PASS

Consider these commands and fill in the correct blank using a RELATIVE pathname. It should be the SHORTEST possible solution.

```
cd /usr/share/perl
cd ../../../etc
pwd
```

Where the "pwd" command prints "/etc"

Check Tests: Ready

Correct cwd PASS

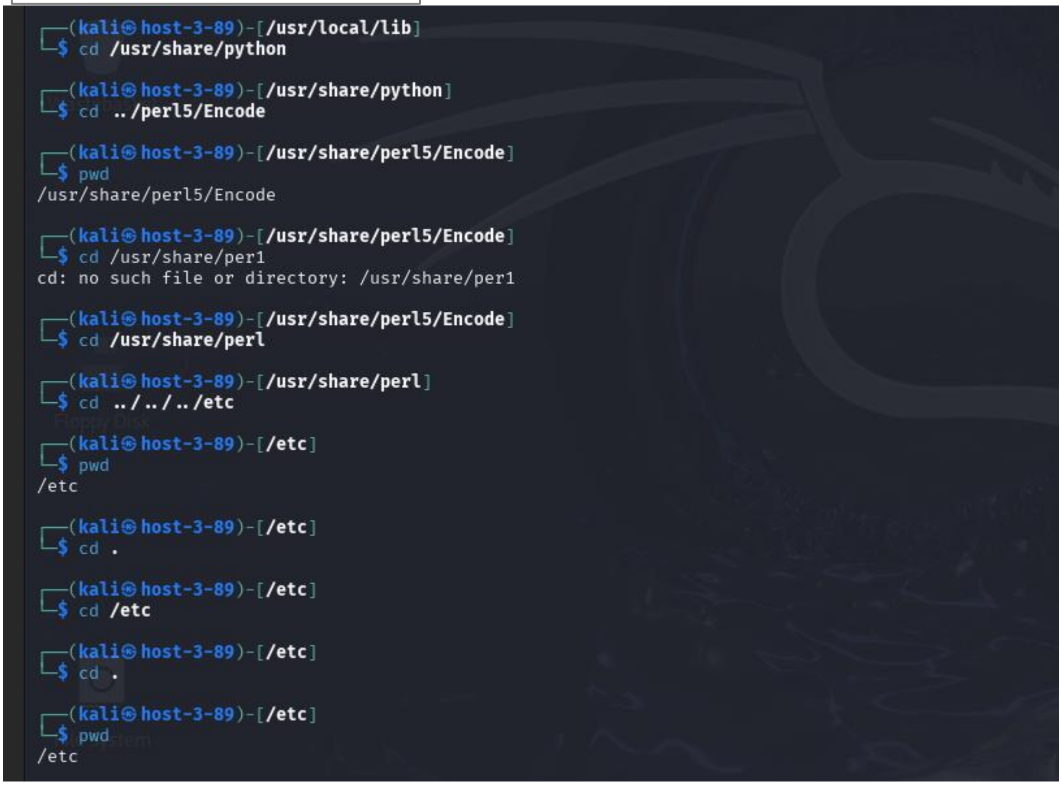
Consider these commands and fill in the correct blank using a RELATIVE pathname. It should be the SHORTEST possible solution.

```
cd /etc
cd .
pwd
```

Where the "pwd" command prints "/etc"

Check Tests: Ready

Correct cwd PASS




# Question 5 Nano Editing

Use nano to create a file /home/kali/edit1. Cut and paste the following text into edit1 and save the file. Remember you cannot easily cut and paste to a vnc terminal, so use ssh. Do not insert additional lines (even blank lines) or extra space characters.

```
asdaslkalsdklnnnne lazy dog quick frog
6f2d9937604b8422abc7493a7ff0c884 /etc/host.conf
This is an exercise!
Up, down,
left, right,
build your terminal's
muscles bit by bit
```

In all the editor questions you must WRITE the file in order to pass the question.

Check Tests: Ready

| | |
|---|---|
| Line 1 found somewhere | PASS |
| Line 2 found somewhere | PASS |
| Line 3 found somewhere | PASS |
| Line 4 found somewhere | PASS |
| Line 5 found somewhere | PASS |
| Line 6 found somewhere | PASS |
| Line 7 found somewhere | PASS |
| All edits complete | PASS |

Delete the word "an" from line 3, plus one of the spaces. The line left should read "This is exercise!".

Check Tests: Ready

line check PASS

Add " and byte by byte" to the end of the line "muscles bit by bit".

Check Tests: Ready

line check PASS

Append to the end of the file a new line which reads:

```
123456789 123456789
```

```
└─$ nano edit1

┌──(kali㉿host-3-89)-[~]
└─$ cat edit1
asdaslkalsdklnnnne lazy dog quick frog
6f2d9937604b8422abc7493a7ff0c884 /etc/host.conf
This is exercise!
Up, down,
left, right,
build your terminal's
muscles bit by bit

┌──(kali㉿host-3-89)-[~]
└─$ nano edit1

┌──(kali㉿host-3-89)-[~]
└─$ cat edit1
asdaslkalsdklnnnne lazy dog quick frog
6f2d9937604b8422abc7493a7ff0c884 /etc/host.conf
This is exercise!
Up, down,
left, right,
build your terminal's
muscles bit by bit and byte by byte

┌──(kali㉿host-3-89)-[~]
└─$ nano edit1

┌──(kali㉿host-3-89)-[~]
└─$ cat edit1
asdaslkalsdklnnnne lazy dog quick frog
6f2d9937604b8422abc7493a7ff0c884 /etc/host.conf
This is exercise!
Up, down,
left, right,
build your terminal's
muscles bit by bit and byte by byte
123456789 123456789
```

Check Tests: Ready

All edits complete PASS

Using mark (^^ i.e. CTRL and ^) mark the whole of the first line of the file and then cut (^K) that line out. Move that line and paste it back in (^U) so that the line is now line 2 in the file. Edit afterwards to make sure there is not a blank line left at the start.

Check Tests: Ready

All edits complete PASS

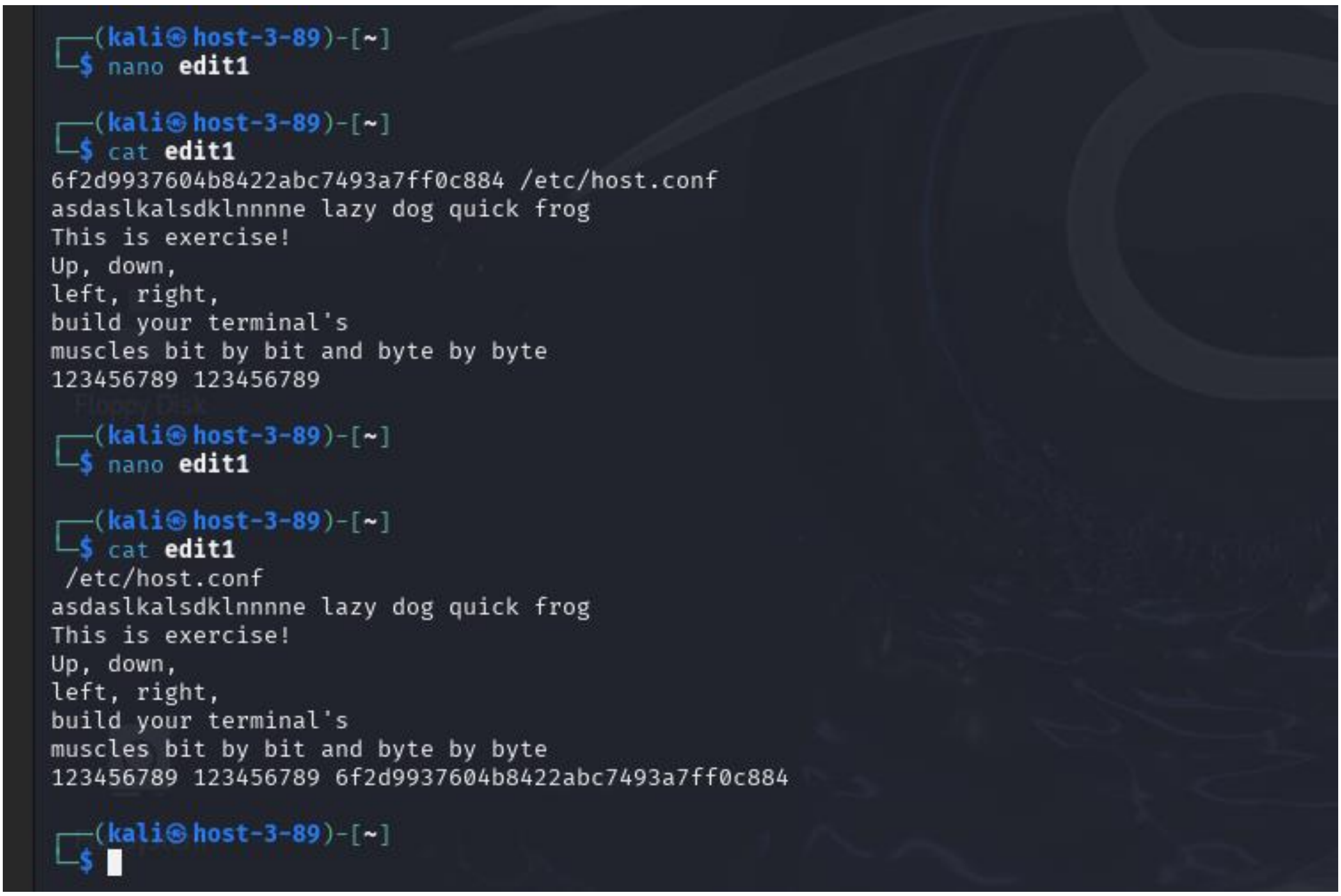
Check Tests: Ready

Now cut out the long hex word on line 1 (beginning 6f2 and ending 884) Leave this first line with a leading space. Now put this hex number at the end of the last line (after 6789) making sure to put a space between the 6789 and the 6f2. Save the file.

Check Tests: Ready

All edits complete PASS

---



```
┌──(kali㉿host-3-89)-[~]
└─$ nano edit1

┌──(kali㉿host-3-89)-[~]
└─$ cat edit1
6f2d9937604b8422abc7493a7ff0c884 /etc/host.conf
asdaslkalsdklnnnne lazy dog quick frog
This is exercise!
Up, down,
left, right,
build your terminal's
muscles bit by bit and byte by byte
123456789 123456789

┌──(kali㉿host-3-89)-[~]
└─$ nano edit1

┌──(kali㉿host-3-89)-[~]
└─$ cat edit1
 /etc/host.conf
asdaslkalsdklnnnne lazy dog quick frog
This is exercise!
Up, down,
left, right,
build your terminal's
muscles bit by bit and byte by byte
123456789 123456789 6f2d9937604b8422abc7493a7ff0c884

┌──(kali㉿host-3-89)-[~]
└─$ 
```